

Activating Metal–Organic Cages by Incorporating Functional $M(\text{ImPhen})_3$ Metalloligands: From Structural Design to Applications

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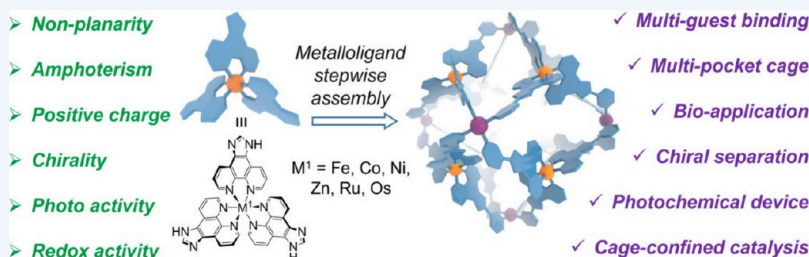
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CONSPECTUS: The emulation of ingenious biofunctions has been a research focus for several decades. Metal–organic cages (MOCs), as a type of discrete supramolecular assembly with well-defined shapes and cavities, have aroused great interest in chemists to imitate natural protein cages or enzymes. However, to genuinely achieve tailored functionalities or reactivities of enzymes, the design of cage structures combining both the confined microenvironment and the active site is a prerequisite. Therefore, the integration of functionalized motifs into MOCs is expected to provide a feasible approach to construct biofunctional confined nanospaces, which not only allows the modulation of cage properties for applications such as molecular recognition, transport, and catalysis but also creates unique microenvironments that promote enzymatic effects for special reactivities and selectivities, thereby providing a versatile platform to achieve exceptional biomimetic functions and beyond.

In this Account, we specifically focus on our research toward engineering active confined-nanospaces in MOCs via incorporation of $M(\text{ImPhen})_3$ metalloligands, a typical tris-chelate coordination moiety comprising imidazophenanthroline ligands and variable metal ions, as the principle functional units for stepwise assembly of active-MOCs. Starting from their structure design and merits, we describe the versatility of $M(\text{ImPhen})_3$ centers for multifunctionalization of the confined cage-nanospaces. By integrating different metal ions like Ru, Os, Fe, Co, Ni, Zn, the metal ion inherent properties, e.g., redox activity of Fe/Co-centers, chirality, and photoactivity of Ru-centers, and dynamics of Co/Zn-centers, could be integrated and tailored on the cages as isostructural nanosized containers or reactors. Changing the Pd or Pt cage vertices to organic clips could remarkably enhance acid–base stability and endow cages with flexibility and allostery. Utilization of ImPhen organic ligands containing imidazole groups introduces proton transfer capability, which can couple with the high-positive charges on the cage to create amphoteric microenvironments in the porous open-cage solution. Moreover, the nonplanar stereoconfiguration of $M(\text{ImPhen})_3$ confers multiple peripheral pockets on the cage, which render multisite, high-order, and dynamics guest binding for the benefit of applications such as drug delivery, molecular separation, and catalytic turnover.

The construction of active-MOCs from tailorable $M(\text{ImPhen})_3$ metalloligands provides us with a new perspective on their structural design and functionalities. Merging the cage confinement with distinct physicochemical properties on a supramolecular level makes it practical to realize synergistic and cooperative effects for functionality enhancement beyond molecular components or the reactivity different from the bulky solution, which could largely expand the potential of MOCs as a multirole platform to wide application scenarios such as artificial photosynthesis, unconventional catalysis, and theranostic nanomedicine.

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- Guo, J.; Fan, Y.-Z.; Lu, Y.-L.; Zheng, S.-P.; Su, C.-Y. Visible-Light Photocatalysis of Asymmetric $[2 + 2]$

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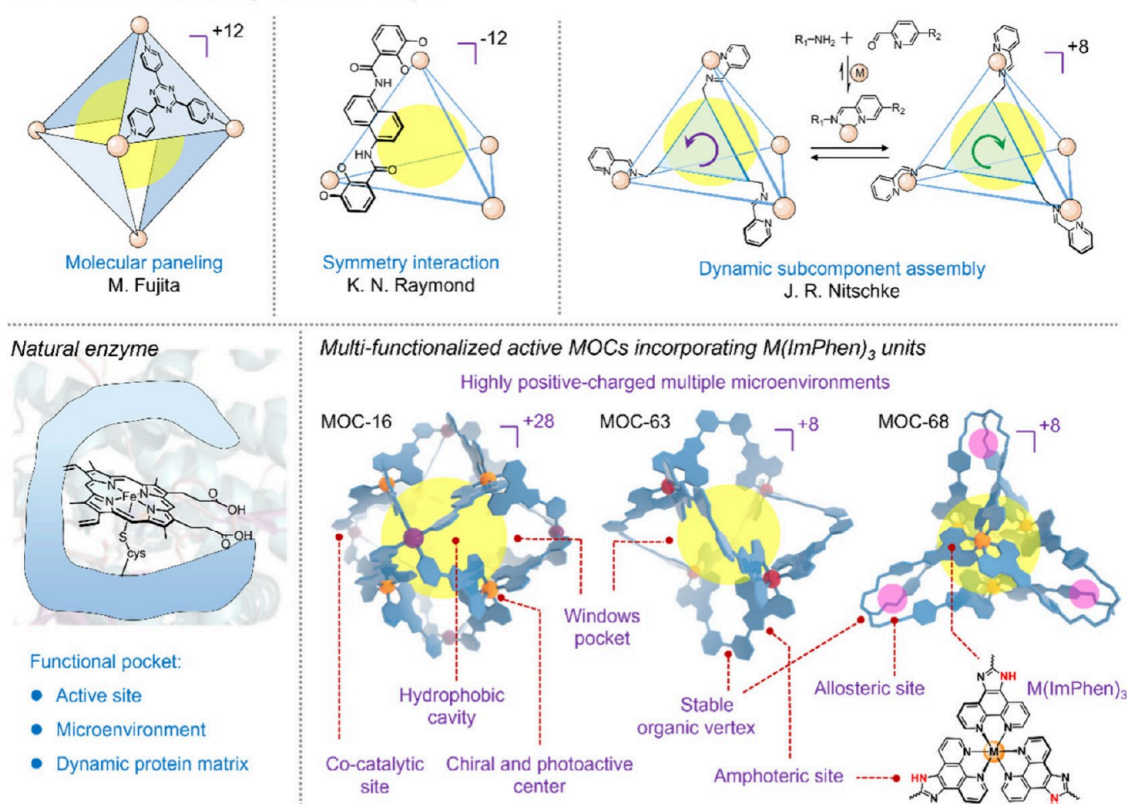
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Scheme 1. Biomimetic MOC Structures and Assembly Strategies

MOCs characteristic of organic faces or edges



Cycloaddition in Cage-Confined Nanospace Merging Chirality with Triplet-State Photosensitization. *Angew. Chem., Int. Ed.* **2020**, 59, 8661–8669.² This work initiated our research on the cage-confined visible light asymmetric photocatalysis on $[2 + 2]$ cycloaddition.

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- Huang, Y. H.; Lu, Y. L.; Ruan, J.; Zheng, S. P.; Zhang, X. D.; Liu, C. H.; Qin, Y. H.; Cao, Z. M.; Jiao, Z.; Xu, H. S.; Su, C. Y. Dynamic Metallocupramolecular Cages Containing 12 Adaptable Pockets for High-Order Guest Binding Beyond Biomimicry. *J. Am. Chem. Soc.* **2023**, 145, 23361–23371.⁴ This work demonstrated the hierarchical and high-order guest binding ability of the adaptive MOC-16-type cage with multiply pockets.

1. INTRODUCTION

The covalent molecular cages possessing inner cavities have long been studied as a paradigm of host–guest interactions in supramolecular chemistry, and in recent decades, the design and self-assembly of supramolecular cages, e.g., coordination-driven assembly of metal–organic cages (MOCs), emerged as a growing topic for biomimicry research.^{5–14} In this field, the structural bionics has been successfully developed through judicious coordination engineering of the cage-confined

nanospaces in MOCs, where the spatial confinement and positive/negative charges reminiscent of protein cages and enzyme pockets are specifically interested (Scheme 1).^{14,15} However, the functional bionics of natural enzymes needs a combination of the active sites with the microenvironment that is maintained by the dynamic protein matrix. To achieve an artificial counterpart capable of rivaling biofunctions, the incorporation of functional units into the host structure may offer a feasible synthetic strategy to activate the cage-nanospaces for enhancement of function and performance. Nevertheless, the synthetic development of real active-MOCs is still a work in progress. While many examples tailored for specific tasks have been reported, the cage models capable of integrating multifunctionalities or installing different metal-centers are scarce. Such cage models may enable more effective and wide applications in catalysis,¹⁶ separation,^{17–19} drug delivery,^{20,21} and smart materials^{22,23} to advance this field forward even beyond biomimicry.

The tris-chelate $M(\text{Phen})_3$ and analogs represent a classic-type of Werner coordination compounds that show fruitful properties in catalysis,^{24,25} sensing,²⁶ electron mediation,²⁷ and bioapplications.^{28,29} The $\text{Ru}(\text{Phen})_3$ -type moieties, known for their desirable photochemical and photophysical properties, have been widely studied in supramolecular systems. These moieties are usually introduced through host–guest interactions^{30,31} or modified on the exterior of cages.^{32,33} However, they are less used as the primary building units in self-assembly protocols.^{1,34,35}

As exemplified by paradigmatic cages of Fujita, Raymond, and Nitschke et al.,^{5,6,36} the classic strategies for the synthesis of MOCs are to utilize linear or polygonal organic ligands as

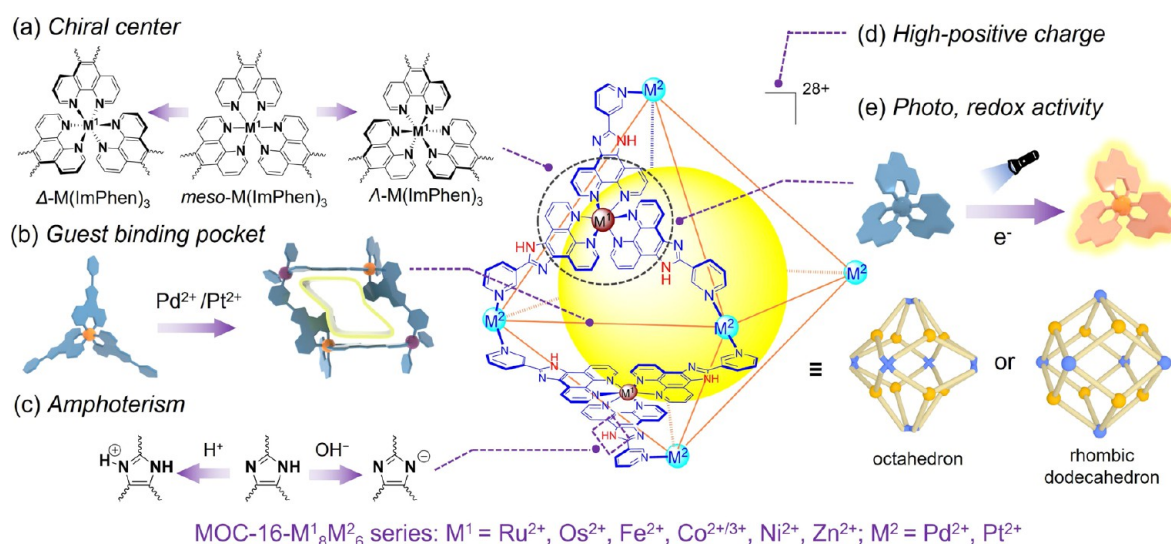


Figure 1. Intrinsic properties brought forth by the M(ImPhen)₃ units in MOC-16 series cages: (a) chiral-on-metal character, (b) windows pocket for dynamic guest binding imparted by the nonplanar fan-like metalloligands, (c) amphoteric property based on proton transfer of imidazole groups for acid–base and charge-state modulation, (d) high-positive charges from metal ions, and (e) photo and redox activities inherent in metal-centers.

the cage edges or faces to assemble directly with metal ions (Scheme 1). The functions of these kinds of MOCs mainly depend on the shape and size of the confined cavities or the limited traits of organic ligands and metal centers. The physiochemical activities of the cage subcomponents have not been fully investigated. We envisage that utilization of M(Phen)₃-type complexes as subcomponents in MOC skeleton would bring their versatile properties into the cage-confined nanospaces, thereby enhancing the functionality and performance. Moreover, the stepwise assembly of bimetallic cages from prepreparation of the metalloligands allows for the integration of both inert Ru/Os and labile Fe/Co/Ni/Zn metal ions into the cage framework, leading to isomerization of cages with diversified properties and dynamics. To further emulate the biofunctions, we also expand Phen ligand into imidazophenanthroline (ImPhen) to introduce imidazole groups, which ubiquitously present in biosystems as a pivotal protein residue. A series of MOC-structures based on M(ImPhen)₃ units have been designed and synthesized in our group (Scheme 1).

These artificial cage models showcase unique appeal not only by imitating enzyme microenvironments for structural bionics but also by integrating the inherent activities of metals or ligands for functional bionics. Based on the choice of metal-centers and organic linkers for cage assembly, a variety of functions could be imparted to the cages via incorporation of chiral-centers, catalytic-centers, photoactive-centers, redox-centers, allosteric-sites, amphoteric-sites, and so on. Moreover, the high-positive charges, hydrophobic cavities, peripheral pockets, framework flexibility, coordination dynamics, etc., could be expected from M(ImPhen)₃ multicomponents (Scheme 1). Significantly, synergistic action and coupling effect would be achieved by installing multifunctional units on the cages, which work with the cage-confinement effect to largely enhance the performance to rival or even surpass that of proteins or enzymes.

In this Account, we focus on how to design M(ImPhen)₃-based active-MOCs to achieve multifunctionalities and their synergism for multirole tasks, covering our 10 years' exploration in their host–guest attributes and applications in

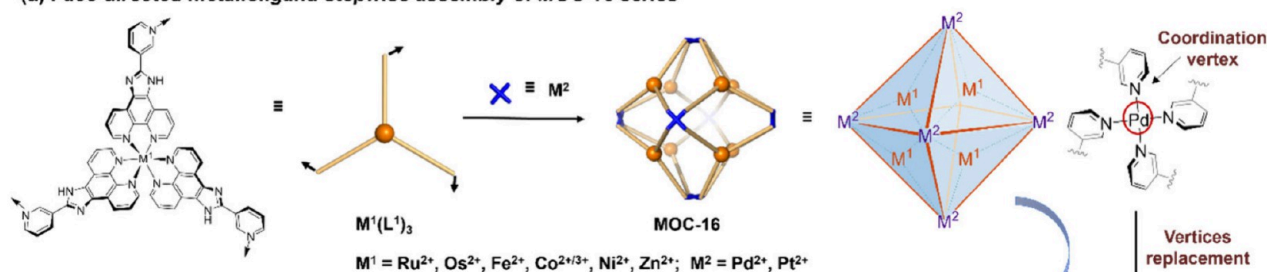
chiral recognition, resolution and separation, photochemical, amphoteric, and allosteric catalysis, as well as drug-delivery, imaging, and theranostic combination. We hope the described results could inspire more research endeavors through a new perspective of MOC design and assembly, not only aiming at biomimetic purpose, but also pushing this field toward wider and practical applications.

2. DESIGN PRINCIPLE AND SYNTHESIS

Our design principle focuses on utilizing M(ImPhen)₃ moieties as the key functional units in the cage assembly. As illustrated in Figure 1, the tris-chelate M(ImPhen)₃ cores can render diversified attributes into MOC-16-type cages:^{1,4,37,38} (a) metal-centered chirality with stereochemical stability depending on metal ion lability, affording Δ-/Λ-enantiomeric or mesomeric cages;^{37–40} (b) multiple rhombic open-pockets arisen from nonplanar stereoconfiguration of M(ImPhen)₃, facilitating high-order and dynamic guest binding and exchange with broad guest adaptability;⁴ (c, d) electrostatic effect of the highly charged positive cage, which combines with protonation–deprotonation behavior of imidazoles to result in amphoteric property, creating microheterogeneity of the porous-cage solution beneficial for unique reactions such as basic catalysis in acid solution, and acid–base dual-role or cascade catalysis;^{41–43} and (e) photo and/or redox activities for UV–visible light harvesting and one-/two-photon excited emission, enabling electron or energy transfer.

The well-ordered disposition of the multiple and distinct functional units within a single-MOC makes it efficient to synergize these functions on a supramolecular level for multirole utilities. For example, installing 8 Ru(ImPhen)₃ photocenters on the cage makes it an effective photochemical molecular device (PMD) for H₂ production,^{44–46} while coupling with the chirality and guest recognition by pockets turns it into a nanoreactors for the cage-confined photocatalysis of unusual reactivity and selectivity.^{2,47,48} Moreover, multirole tasks could be implemented by a single-cage via its multifunctionalities, e.g., combined drug transport, imaging and therapy.^{49–51}

(a) Face-directed metalloligand stepwise assembly of MOC-16 series



(b) Vertex-directed organic-clip chelation assembly of MOC-63



(c) Post-synthetic metalloligand stepwise assembly of MOC-68

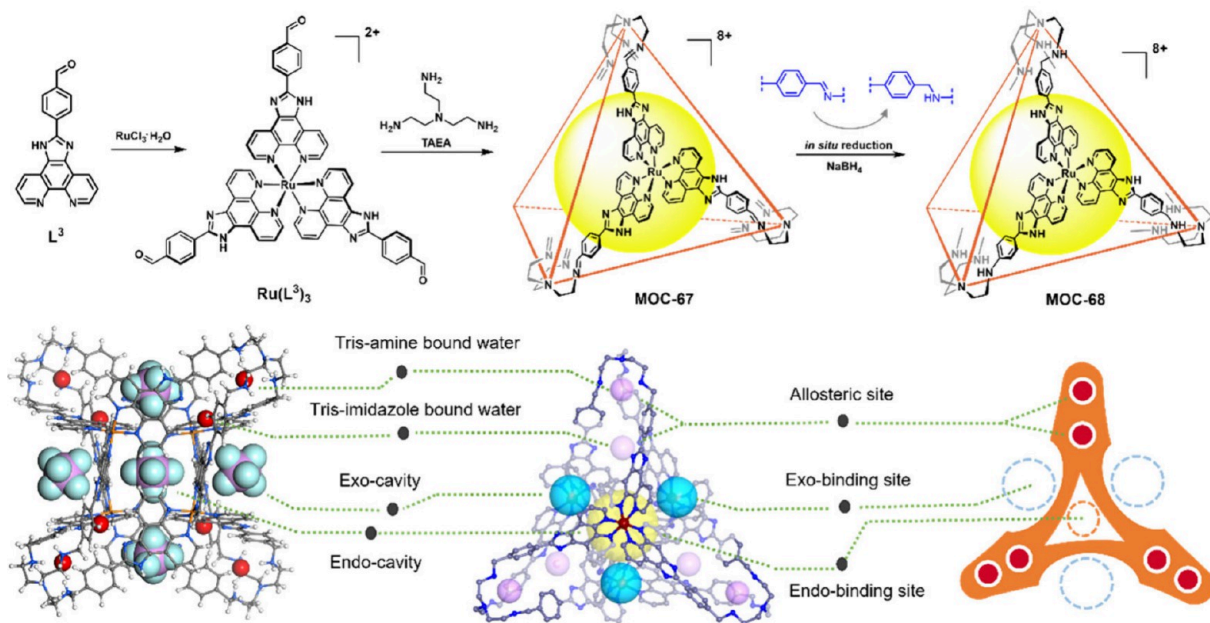


Figure 2. Three assembly strategies of active-MOCs. (a) Heterometallic MOC-16 cage series involving different metal-centers. (b) Acid–base dependent charge tunable redox MOC-63. (c) Protein-like allosteric MOC-68. Adapted with modifications from ref 52. Copyright 2022 American Chemical Society (ACS), ref 3. Copyright 2023 the authors. Published by Elsevier under a CC BY-NC-ND 4.0 license.

2.1. Synthesis of MOC-16 Series Cages

The heterometallic MOC-16 cage series $[(M^1L^1)_3M^2_6]^{28+}$ ($M^1 = Ru, Os, Fe, Co, Ni, Zn$; $M^2 = Pd, Pt$) are assembled via a face-directed metalloligand stepwise strategy (Figure 2a). In the first step, thermodynamically stable $M^1(ImPhen)_3$ metalloligands are formed owing to the tris-chelate effect, leading to the creation of 6-coordinated M^1 -centers. In the second step, these metalloligands are assembled with 4-connecting mono-

dentate M^2 (Pd^{2+} or Pt^{2+}) to form the cage structure. The resulting cage shows a shape of octahedron with 8 metalloligands as faces and 6 M^2 as vertices. A better description of this type of cage is rhombic dodecahedron, which regards 8 M^1 and 6 M^2 as the vertices and 12 rhombic window-portals as the faces (Figure 1), and exactly denotes 12 cage-faces as peripheral pockets for guest binding except the inner cavity.¹ In general, total 14 divalent metal-centers provide highly

positive charges up to +28, which endow the cage with excellent solubility in aqueous solvents.⁴¹ Noticeably, the stereoconfigurations of $M^1(\text{ImPhen})_3$ impart specific attribute to the multiple open-pockets, which exhibit bowl-shape with two ImPhen ligands in parallel suitable for π - π stacking of guests, and the other two ImPhen ligands in bent arrangement adaptable for guest and substrate variation.^{4,48} The stereochemistry of the cage is also inherited by the pockets, keeping consistent chirality.³⁷ Furthermore, alternation of M^1 -centers leads to isomeric cages with distinct properties, importing variable functionalities or tuning the flexibility and stereodynamics of the cages.^{37,40}

2.2. Synthesis of MOC-63

To improve the cage stability against harsh conditions like acidic, alkaline, or reductive media, an intuitive idea is to replace $\text{Pd}^{2+}/\text{Pt}^{2+}$ vertices in MOC-16 with robust linkers to assemble the $M(\text{ImPhen})_3$ units. Therefore, we design ditopic ligand L^2 as a molecular clip to accomplish vertex-directed organic-clip chelation assembly of MOC-63 (Figure 2b).⁵² MOC-63 resembles Fujita's octahedral Pd_6L_4 cage with 4 faces open to allow guests capture into the central cavity. The covalent vertices endow the cage with excellent tolerance toward strong acid and bases, while the tris-chelate $\text{Fe}(\text{ImPhen})_3$ based on the labile Fe-center persists local dynamics, imparting redox activity to the cage for catalysis.⁵² Similar to the electrostatic effect of proteins, the surface charge-state largely affects the polarity and solubility of the cage. Total 12 imidazole groups endow the cage with a wide range of charge tunability from -4 to $> +8$ through protonation-deprotonation equilibria to counteract metal ion charges, which can modulate the cage solubility to benefit recovery and recycling use of MOC-63 as a catalyst.

2.3. Synthesis of MOC-68

In order to mimic allosteric property of natural proteins, we introduce tritopic covalent linker to implement postsynthetic metalloligand stepwise assembly of MOC-68 (Figure 2c).³ This strategy employs dynamic imine covalent bonds to assemble $\text{Ru}(\text{ImPhen})_3$ units, leading to MOC-67 which is subsequently reduced to highly robust metalloamine cage MOC-68.

MOC-68 is a tetrahedral-shaped cage consisting of 4 flexible amine-vertices and 4 nonplanar $\text{Ru}(\text{ImPhen})_3$ subcomponents, which forms 6 open exorhomboidal pockets, 1 central endocavity, and 8 water-binding allosteric sites. This cage features in remarkable protein-resembling characters: (1) multiple imidazoles and amines for protonation, deprotonation and water binding; (2) a wide range charge-states adjustable by pH-variations; (3) distinct exo- and endocavities for guest accommodation and exchange; and specifically (4) proton-driven conformation allostery for multistate guest binding switch that imitates the omnipresent allosteric proteins.³

2.4. Chirality Control of MOC-16 Cages

The chiral-at-metal stereochemistry of $M(\text{Phen})_3$ confers the metal-centers on the cage with intrinsic Δ/Λ -chirality. Owing to cooperative stereochemical coupling and the self-sorting effect during assembly, a racemic mixture of homochiral MOC-16 cages with Δ_8 - and Λ_8 -configurations is formed simultaneously in the solution. To resolve cage enantiomers, we developed two chiral cage synthetic methods based on inert Ru-centers and metastable/labile Fe/Co/Ni/Zn-centers (Figure 3).

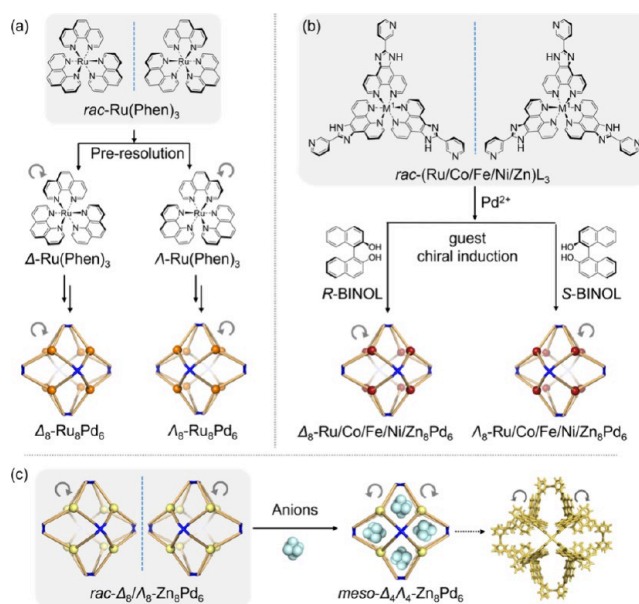


Figure 3. Chiral control of MOC-16-type cages. (a) Preresolution of inert Ru-precursors and (b) chiral-induction of metastable metal-centers in case of dual-guest binding mode. (c) Anion-induced mesomerization of MOC-16- Zn_8Pd_6 . Adapted with modifications from ref 37. Copyright 2023 Wiley-VCH.

For the stereochemically stable Ru-cages, preresolution of $\text{rac-Ru}(\text{Phen})_3^{2+}$ precursor by $\text{K}_2[\text{Sb}_2(+)\text{-tartrate}]_2 \cdot 3\text{H}_2\text{O}$ chiral-induction agent enables preparation of Δ -/ Λ - RuL_3 metalloligands with desired enantiopurity via oxidation and Debus-Radziszewski imidazole synthesis reactions. The coordination assembly of Δ -/ Λ - RuL_3 enantiomers with $\text{Pd}^{2+}/\text{Pt}^{2+}$ gives rise to enantiopure Δ_8 -/ Λ_8 -MOC-16 cages.³⁹ In contrast, the stereolability of first-row transition metal ions causes racemization/epimerization problems that pose a challenge to directly synthesize the enantiomeric cages of MOC-16-Fe/Co/Ni/ Zn_8Pd_6 . Instead, the guest chiral-induction methods are applied by using R- and S-BINOLs as inducers, which play a vital role either through an in situ assembly process to directly induce formation of Δ_8 -/ Λ_8 -MOC-16-Fe/Co/ Zn_8Pd_6 cages with labile metal-centers, or through a postassembly process to cocrystallize Δ_8 -/ Λ_8 -MOC-16 cages of Ru/Ni/Fe/Co/ Zn -centers (Figure 3b).^{37–40}

It is noteworthy that the cage stabilization effect could effectively consolidate stereodynamics of labile metal-centers, such as Zn^{II} and Co^{II} , through the mechanical coupling; however, cage mesomerization is able to be induced and fixed by PF_6^- , SbF_6^- , or ClO_4^- anions, which gives *meso*- $\Delta_4\Lambda_4$ -MOC-16-Co/ Zn_8Pd_6 cages as intermediate state for transformation between homochiral Δ_8 - and Λ_8 -MOC-16-Co/ Zn_8Pd_6 cages (Figure 3c, see supplemental discussion).^{37,38}

3. HOST-GUEST CHEMISTRY AND PROPERTY

3.1. High-Order Guest Binding Capacity

The guest-binding >4 in a single supramolecular cage remains rare and the quantitative characterization is challenging yet. The multiportal feature of MOC-16-type cages arisen from fan-like configurations of $M(\text{ImPhen})_3$ metalloligands offers ideal models to achieve high-order and hierarchical binding of guests. Solution ^1H NMR and ESI-MS studies reveal dynamic host-guest interactions and a broad binding scenarios up to

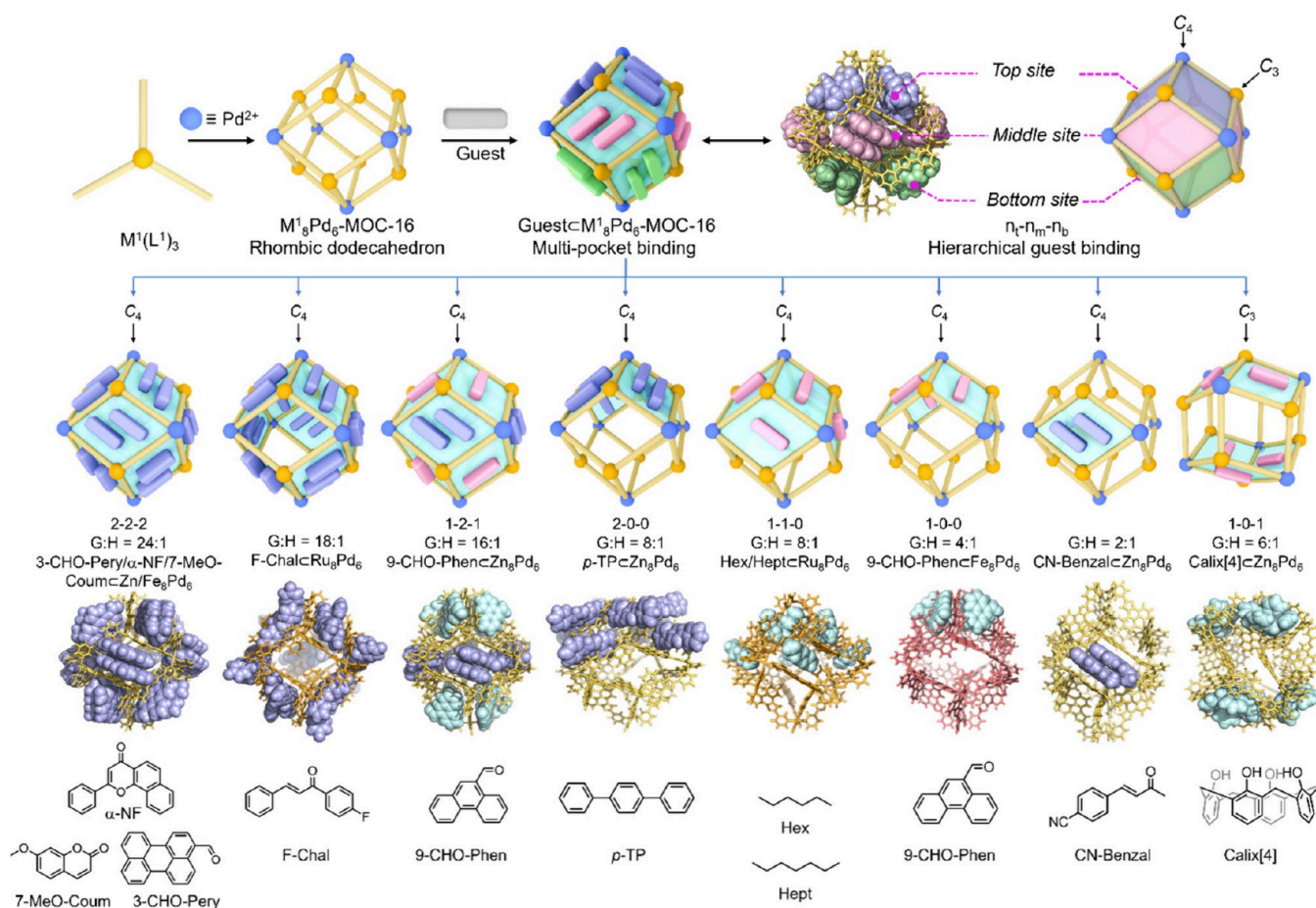


Figure 4. High-order and hierarchical multiguest binding models of MOC-16-type cage. Adapted with modifications from ref 4. Copyright 2023 ACS.

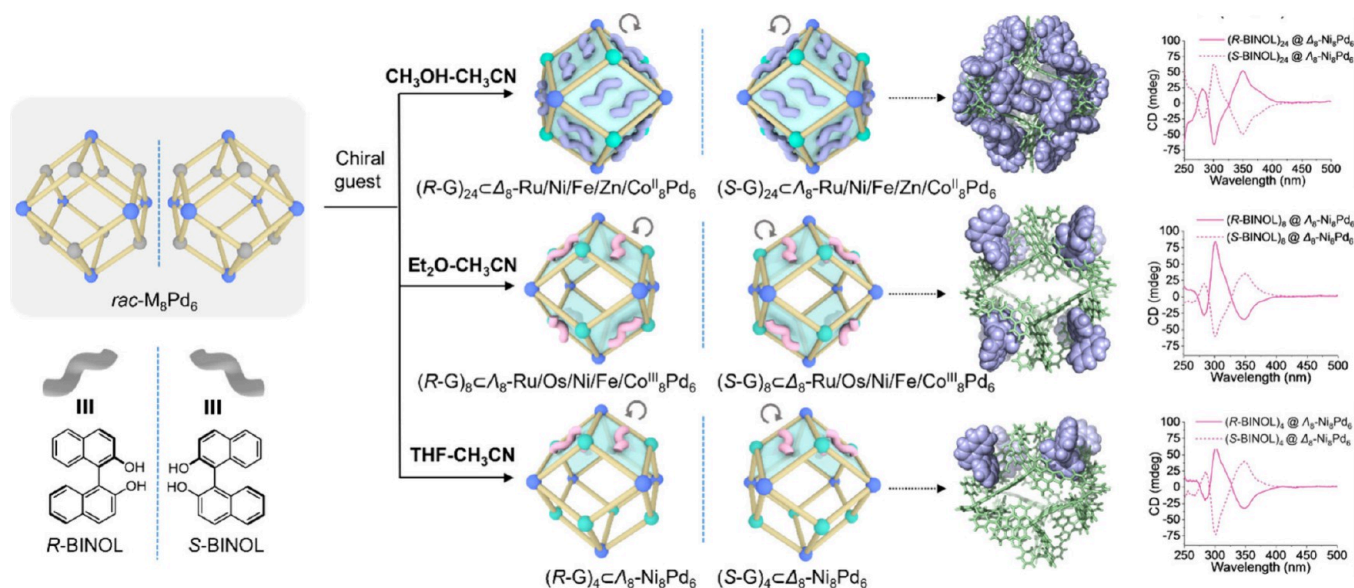


Figure 5. Chiral-guest recognition models of MOC-16-type cage showing characteristic CD patterns. G = guest, H = host-cage; R = R-BINOL, S = S-BINOL. Adapted with modifications from ref 37. Copyright 2023 Wiley-VCH.

dozens of guests, while solid-state single-crystal X-ray crystallography provides a clearer view of the multisite binding repertoire of multiple guests, which is rarely found in other MOCs.^{4,15,48} As illustrated in Figure 4, the 12 cage-pockets

located on 12 open-faces of the rhombic dodecahedron can be divided into three levels ($n_t-n_m-n_b$ for top-middle-bottom guest-binding numbers), leading to variable multiguest binding events with captured guests ranging from 2 to 24. The

multipockets work separately but in synergistic action, accommodating guests either in dual-guest packing or in single-guest binding modes. It is worth mentioning that the lability of the metal ion centers plays a significant role in imparting dynamism to the whole cage, and the dynamic Zn-centers enable MOC-16- Zn_8Pd_6 to bind the oversized Calix[4] guests. Therefore, an extended dynamic-fit multiguest binding mechanism is established to cover lock-and-key, induced-fit, conformational selection, and mutual induced-fit phenomena beyond known protein interacting models, which lays the foundation of applications in cage catalysis, separation, and drug delivery.

3.2. Chiral-Guest Recognition

The variable multiguest binding behaviors of enantiopure homochiral Δ_8 -/ Λ_8 -MOC-16 cages result in intrigue chiral-guest recognition and discrimination ability depending on the host–guest adaptability (Figure 5).^{4,37,40} The dynamic host–guest interactions are found to be profoundly influenced by the cage flexibility, guest solubility, and solvent media, giving rise to two types of reverse diastereoselective recognition processes subject to guest encapsulation capacity in one pocket. In general, the dual-guest binding mode leads to *R*-G Δ -H and *S*-G Δ -H enantiomeric pairs in CH_3OH – CH_3CN solvent, whereas under Et_2O – CH_3CN and THF – CH_3CN conditions, only partial pockets are occupied to give opposite *R*-G Δ -H and *S*-G Δ -H pairs in single-guest binding mode.

3.3. Microheterogeneity in Porous Open-Cage Solution

Distinct from the transient solvent cage effect, the thermodynamically stable coordination cages are able to create the cage-confined nanospaces in homogeneous solution to show local microheterogeneity, making the microenvironments in cage-cavities different from the bulk solution akin to natural enzyme pockets (Figure 6).⁴¹ The open-cage windows allow

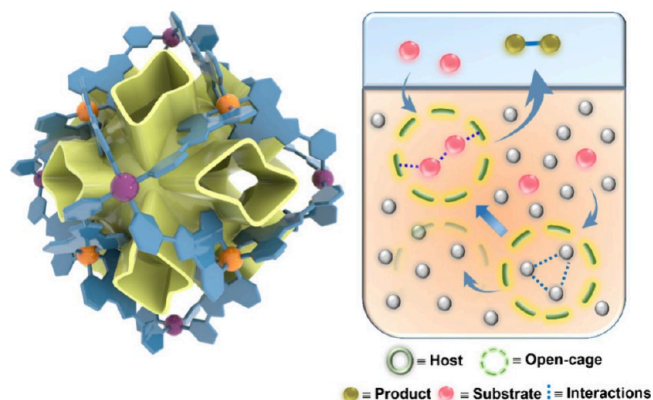


Figure 6. Illustration of the open-cage model of MOC-16 and the cage-confined nanospaces in solution for guest exchange. Adapted with permission from ref 41. Copyright 2021 the authors. Published by Oxford University Press under CC BY 4.0 license.

selective ingress-and-egress of guests driven by recognition and solvophobic effect,¹⁵ which also enables phase-transfer of insoluble guests into the porous cage solution to increase solubility.⁵³ For example, the high +28 charges on MOC-16- Ru_8Pd_6 drives partial deprotonation of the 24 imidazole-NH groups, shifting proton transfer equilibrium and lowering $\text{p}K_a$, which generates Brønsted acidity-basicity outside-and-inside the cage, respectively. This amphoteric property opens a

special avenue for acid–base dual-role or cascade catalysis in solution.^{36,41–43}

3.4. Allosteric Host–Guest Regulation

Allostery is a biological phenomenon of the regulation of protein function, structure, or flexibility through the binding of effector at distant site, which is omnipresent in nature often referred to as the “second secret of life”.⁵⁴ The robust protein-resembling MOC-68 presents a simple cage model of biomimetic allostery with water as effector.³ The flexible triamine-vertices, combined with multiple imidazole-sites for water binding, allow the cage to undergo proton-driven transformation between relaxed-state (R-state) and tensed-state (T-state) to regulate the host–guest dynamics and motions (Figure 7). The exo- and endobinding guests can be

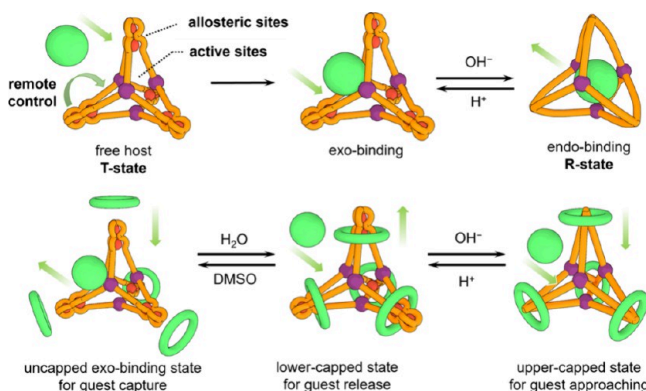


Figure 7. Allosteric host–guest dynamics controlled by remote H_2O -binding, and capping-and-uncapping ring-on-cage motion. Adapted with permission from ref 3. Copyright 2023 the authors. Published by Elsevier under CC BY-NC-ND 4.0 license.

modulated not only via pH-variation but also via secondary guest-binding, motivating guest shift between exobinding and endobinding states via the transition from R-state to T-state. A ring-on-cage system $(\text{CB}[10])_4@ \text{MOC-68}$ is also developed via binding of $\text{CB}[10]$ rings on the vertices. Intermolecular interactions for the ring-on-cage motion can be precisely and reversibly modulated by switching multivalence states of MOC-68 through pH- or solvent-variations. Similar allosteric effect is observed with anions acting as cofactors and can be applied in manipulating the [2 + 2] photocycloaddition reactions (see supplemental discussion).⁵⁵ Therefore, this biomimetic allosteric cage provides a unique model of multimodule host–guest systems and smart materials, promising for drug delivery and allosteric catalysis.

4. APPLICATIONS

The stereochemistry, optoelectronic properties, multiguest binding, and host–guest dynamics inherited or derived from well-organized $\text{M}(\text{ImPhen})_3$ functional units are expected to largely expand the applications of the activated MOCs, which arises prominent supramolecular cage effects to synergize and enhance multifunctionalities within the cage-confined nanospaces.

4.1. Enantioselective Separation

The molecular separation by using MOCs as porous molecular materials is emerging as a new field taking advantage of cage solubility and processability.¹⁹ The high stereostability and guest-binding capacity endow MOC-16-type cages with great

potential in chiral recognition and separation. The enantiomeric Δ -/ Λ -MOC-16- Ru_8Pd_6 cages show excellent stereoselectivity for atropisomeric compounds possessing C_2 -symmetry.³⁹ Through liquid–liquid extraction processes, the water-soluble cage could selectively enantioseparate chiral organic molecules, achieving 62% *ee* value for 6-Br-BINOL as an example. Variable temperature ^1H NMR studies reveal that the kinetically guest exchange results in an unprecedented dynamic resolution process, due to differential exchange rates and energy barriers between the diastereomeric host–guest complexes (Figure 8). This enantioresolution process is

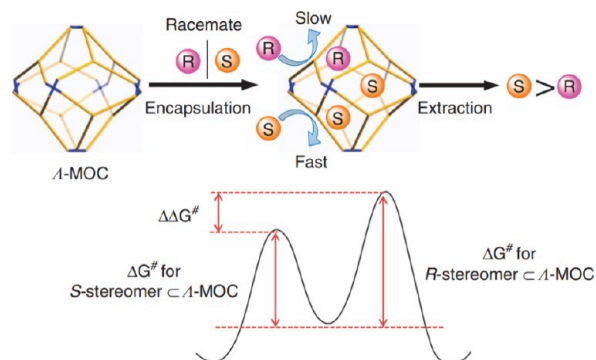


Figure 8. Enantioseparation mechanism of the resolution process. Adapted with permission from ref 39. Copyright 2016 the authors. Published by Springer Nature under CC BY 4.0 license.

applied with more accessible Δ -/ Λ -MOC-16- Fe_8Pd_6 (MOC-42) for a test of stereoselective separation abilities of atropisomers.⁴⁰ The cyclic experiments demonstrate remarkably improved efficiency above 84% *ee* value in 2–4 recycles (Table 1), showing only minimal changes in CD signals of Δ -/ Λ -MOCs-42 after chiral resolution.

Table 1. Enantioselective Separation of Δ -/ Λ -MOC-42

Compounds	Δ -MOC-42		Λ -MOC-42	
	R/S ratio	<i>ee</i> (%)	R/S ratio	<i>ee</i> (%)
<i>rac</i> -BINOL-cycle 1	65:35	30	35:65	−30
<i>rac</i> -BINOL-cycle 4	93:7	86	8:92	−84
<i>rac</i> -6-Br-BINOL-cycle 1	80:20	60	17:83	−66
<i>rac</i> -6-Br-BINOL-cycle 2	90:10	80	6:94	−88

4.2. Photochemical Devices

The exceptional optoelectronic properties of $\text{Ru}(\text{ImPhen})_3$ (visible-light LMCT absorption, long-lived triplet-state, electron/energy transfer, oxidative/reductive quenching processes, etc.) render fruitful photochemical applications to MOC-16- Ru_8Pd_6 cages. The initial study of this cage in 2014¹ demonstrates a cage photoshielding effect for the photosensitive molecules, which inspires our later research to use this photoactive cage as photochemical devices, drug delivery transporters, photocatalytic nanoreactors, and theranostic nanomedicine.

Applying MOC-16- Ru_8Pd_6 as a luminescent host to encapsulate water-insoluble Fc derivatives or TTF molecules, photoluminescence (PL) signals of the cage can be switched and modulated through both chemical or photochemical pathways. This enables achievement of higher-level logic devices with complex and integrated functions.⁵⁶ For example, enclathration of TTF guest into cage-pockets causes PL

quenching of MOC-16, which can be reversibly controlled by stepwise oxidation or reduction among TTF, TTF^{•+}, and TTF²⁺, involving photoinduced electron-transfer (PET) or fluorescence resonance energy transfer (FRET) processes. These two quenching protocols can effectively simulate the simple reversible off/on logic gate or memristor-like sequential memory and Boolean operation functions, providing the possibility to develop more advanced multifunctional artificial systems that mimic single-molecular microdevice.

As a photochemical molecular device (PMD) integrating multiple chromophoric Ru-photosensitizers, catalytic Pd-centers and electron relay components into a single-cage, MOC-16- Ru_8Pd_6 enables highly efficient intramolecular PET to promote photoreactions. For instance, effective visible-light-driven photocatalytic H_2 -production requires the transfer and accumulation of multiple electrons from the light-harvesting centers to the catalytic sites. The highly symmetric MOC-16- Ru_8Pd_6 cage composed of 8 photocenters and 6 catalytic-centers is beneficial for directional PET process to efficiently utilize energy and achieve high turnover numbers (TONs).⁴⁵ Photocatalytic experiments reveal excellent H_2 -evolution activity with a TON of 635 after 48 h and a high initial reaction rate of $380 \mu\text{mol h}^{-1}$. The well-organized photo- and catalytic-centers on the O -symmetric cage offer multiple channels for convergent stepwise PET to enhance H_2 -production (Figure 9). The mechanical coupling of the multicenters increases the cage robustness compared with individual Ru-metalloligands, affording a promising PMD cage-model in homogeneous photocatalytic applications.

Taking advantage of the valid PET process between TTF and MOC-16- Ru_8Pd_6 , the appropriate redox-active TTF guest can act as additional electron-relay mediator to significantly increase the rate of H_2 -evolution.⁴⁴ In contrast, the bulky TTF-derivatives mismatching with the cage-cavity result in interference with the cage PET process, leading to inefficient H_2 -evolution (Figure 10). The mechanism investigation reveals a redox coupling effect between redox TTF and photoredox MOC-16 in the cage-confined chemical nanospace, providing synergistic interplay of photoredox cycles through guest-mediated interactions and communications.

Furthermore, heterogenization of MOC-16- Ru_8Pd_6 cages into ZIF-8, followed by in situ transforming into a carbonate matrix of $\text{Zn}_x(\text{MeIm})_x(\text{CO}_3)_x$ (CZIF), could generate highly durable materials to improve H_2 -production performance (Figure 11). The resulting hybrid material MOC-16@CZIF exhibits enhanced H_2 -evolution activity with a turnover frequency of approximate $0.4 \text{ H}_2\text{s}^{-1}$, which is also proved more resilient under harsh reaction conditions.⁴⁶ The structural heterogeneity and synergistic effect render significant elongation of lifetime on excited-state of MOC-16 to microsecond order of magnitude, which is beneficial to the photocatalytic process.

4.3. Cage-Confined or Enhanced Catalysis

Creation of porous and dynamic nanospaces in open-cage solution alike heterogeneous microenvironment in enzyme provides a versatile multirole platform for catalysis. The cage-confined catalysis differs significantly from catalysis in bulk solution due to a combination of various supramolecular effects, including confinement, encapsulation, concentration, stabilization, electrostatic, hydrophobic effect, etc. Together with noncovalent host–guest interactions that facilitate substrate binding and preorganization, efficient electron/

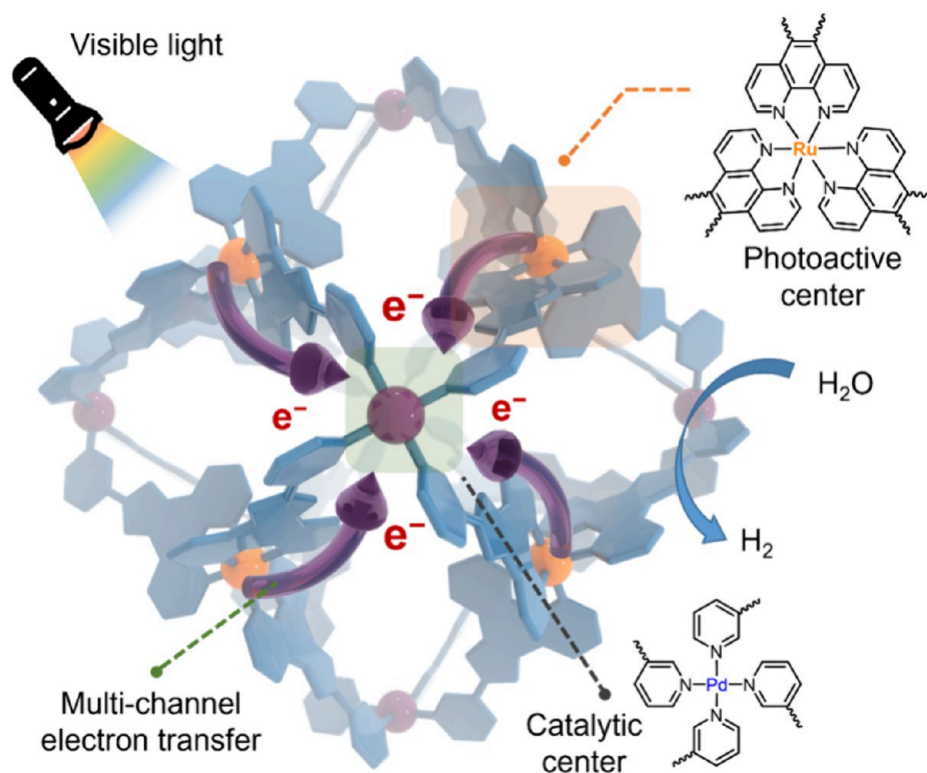


Figure 9. Multichannel ET cage model of PMD for photocatalytic hydrogen production.

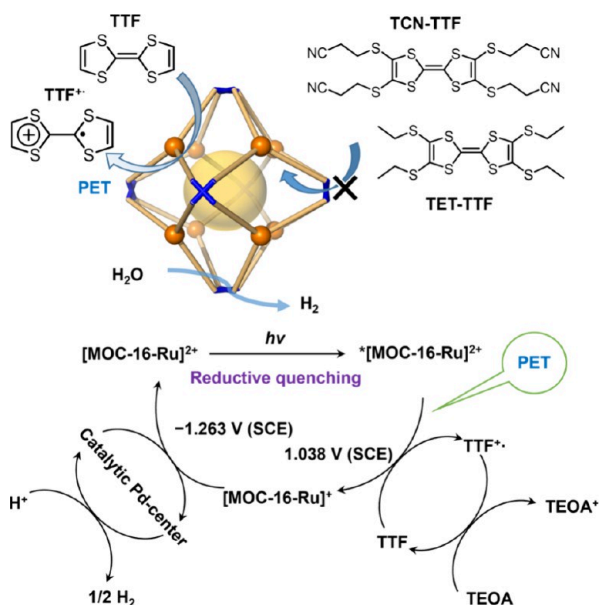


Figure 10. Photoredox cycle coupling with PET processes modulated by redox-active TTF guest as electron-relay in H_2 production. Adapted with modifications from ref 44. Copyright 2019 Wiley-VCH.

energy transfer, and amphoteric microheterogeneity in the cage-confined nanospaces, these effects can generate unconventional reactivities and selectivities for organic transformations.

4.3.1. Amphoteric Catalysis. The charged environment has been demonstrated to play exceptional role in cage-catalyzed reaction. As exemplified by the anionic cage of Raymond et al., the cage stabilizes the cationic intermediates inside the cavity, and the orthoformate hydrolysis, an acid-

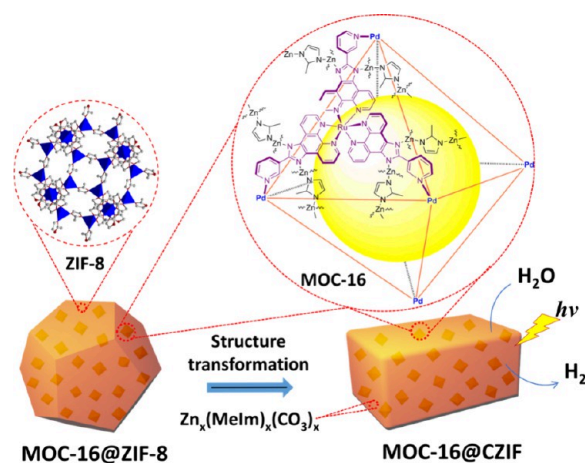


Figure 11. Heterogenizing MOC-16- Ru_8Pd_6 into ZIF-8 followed by transformation to carbonate MOC-16@CZIF for enhanced H_2 evolution. Adapted with permission from ref 46. Copyright 2019 ACS.

catalyzed reaction, can be effectively promoted even in a basic solution.³⁶ We find that the porous open-cage solution of the highly charged cationic MOC-16- Ru_8Pd_6 mediates an amphoteric system suitable for multipurpose acid–base catalysis.^{41–43} The weak $\text{C}_{\text{sp}}-\text{H}$ acidity of phenylacetylene is increased when encapsulated in the highly positive cage-space, thereby accelerating the H/D-exchange process even under acidic conditions (Figure 12).⁴² The successful deuteration of water and base-sensitive alkynes, which is difficult with the conventional method, demonstrates the cage protection effect in the hydrophobic cavity. The one-pot acid/base-catalyzed cascade reactions of acetal derivatives and malononitrile prove the intrinsic-basicity and extrinsic-acidity of the cage solution.⁴¹

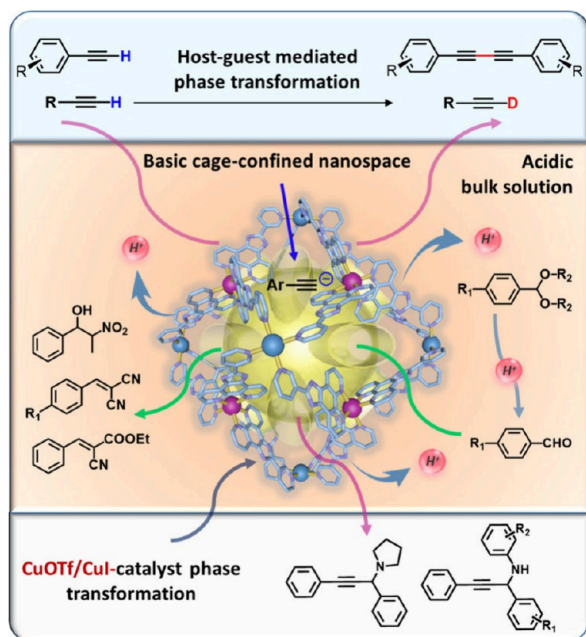


Figure 12. Amphoteric catalysis mediated by basic cage-confined nanospaces in acidic bulk solution, involving the phase transfer of insoluble substrate, product or cocatalyst. Adapted with permission from ref 41. Copyright 2021 the authors. Published by Oxford University Press under a CC BY 4.0 license.

Furthermore, the coordinative ability of imidazole group facilitates loading and transferring insoluble and unstable Cu-cocatalyst onto the cage in aqueous solution, enabling three-component A_3 -coupling reactions in mild solution conditions with up to 71-fold rate enhancement. The base-catalyzed Glaser coupling of terminal alkynes is achieved in acidic $\text{CuCl}@MOC-16$ solution, and the acceleration effect reaches up to 254-fold.^{41,42}

Applying the amphoteric catalysis to asymmetric Diels–Alder (D–A) cycloaddition using chiral Λ -MOC-16- Ru_8Pd_6 successfully yields enantioselective $[4 + 2]$ cycloadducts from both acid-favored anthracene and base-favored 9-anthrone derivatives, reaching up to 96% ee. This demonstrates that the dual-role cage-reactor not only promotes efficient substrate conversion but also has strong chiral control over the bimolecular reactions (Figure 13).⁵⁷

4.3.2. Redox Catalysis. MOC-63 integrating 12 imidazole proton donor–acceptors and 4 redox-active Fe-centers features excellent acid, base, and redox robustness by virtue of cage-stabilized dynamics in solution,⁵² allowing for partial dissociation of $\text{Fe–N}\cdots\text{N}$ chelation bonds while maintaining the cage shape. This dynamic attribute forms the foundation for metal-catalyzed oxidative dehydrogenation reactions. In the confined space, the synergistic action of multiple Fe-centers produces a supramolecular cage effect with a high density of redox-active sites and concentrated reactive intermediates, accelerating multistep radical reactions of tetrahydroquinoline (Figure 14). The acid–base amphoteric imidazole-units make it easily to regulate the solubility of cage via protonation state alteration, enforcing acceleration in acidic condition and recycle of cage-catalyst simply by pH-switching of reaction solution.

4.3.3. Electron/Energy Transfer Photocatalysis. As the homochiral cage imposes the same chirality onto multipockets and $\text{Ru}(\text{ImPhen})_3$ metalloligands bring in versatile photo-

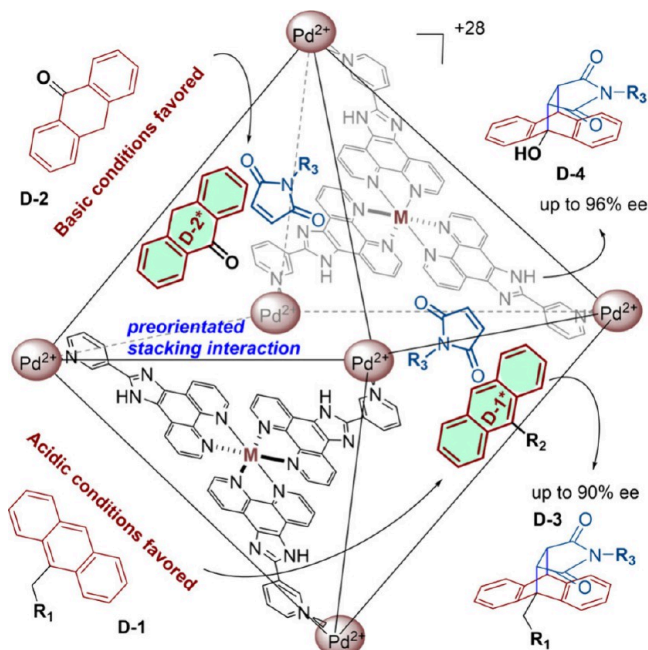


Figure 13. Asymmetric D–A cycloaddition for both acid- and base-conditions favored substrates catalyzed by amphoteric Λ -MOC-16- Ru_8Pd_6 . Adapted with modifications from ref 57. Copyright 2024 ACS.

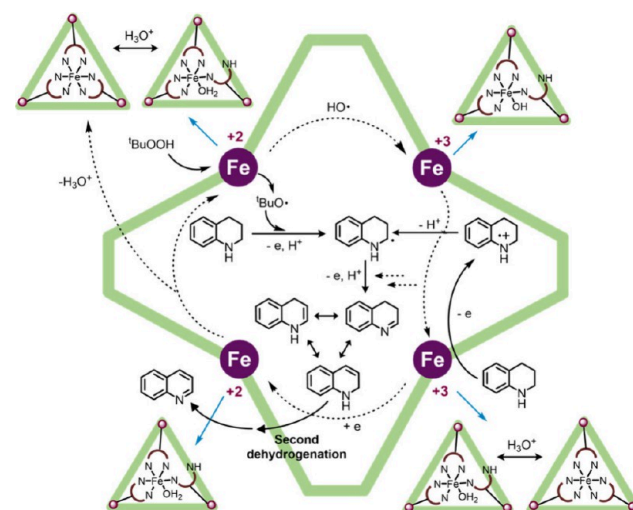


Figure 14. Cage-confined redox catalysis of the multistep radical reactions. Adapted with modifications from ref 52. Copyright 2022 ACS.

chemical and photophysical activities, the Ru-cages represent ideal photoreactor models for asymmetric photocatalysis. The general merits of our photoactive cage-reactors include (1) green and mild conditions of visible-light induced energy transfer (EnT) or electron transfer (ET),⁵⁸ aqueous media, and room temperature, (2) favorite bimolecular recognition, inclusion and preorientation of substrates to control both ground and excited states, and (3) multiply open-pockets favoring dynamic substrate and product exchange, avoiding product inhibition and allowing wide substrate scope. Our work is dedicated to utilizing these enzyme-mimicking nanoreactors to achieve challenging photochemical reactivities at the chemo-, regio-, diastereo-, and enantioselective levels.

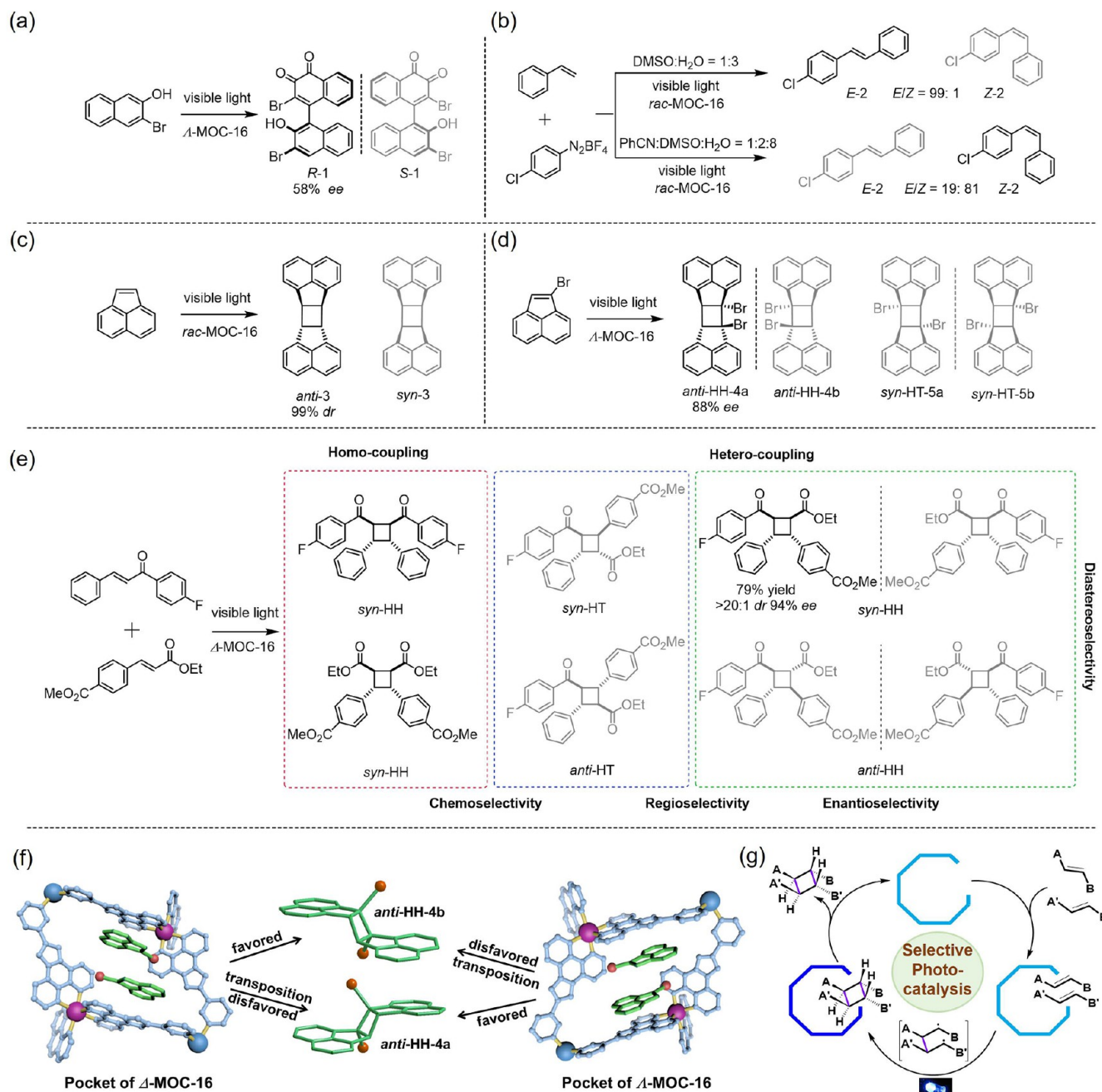


Figure 15. MOC-16- Ru_8Pd_6 catalyzed photoreactions. (a) Regio- and enantioselective 1,4-coupling of 3-bromo-2-naphthol. (b) *E/Z*-selective Meerwein arylation. (c, d) Stereoselective [2 + 2] cycloaddition of acenaphthylene derivatives. (e) Chemo- and stereoselective homo- or heterocoupling of α,β -unsaturated carbonyl olefins. (f) Selectivity control of photodimerization in a chiral pocket via preorganization of two substrates. Adapted with permission from ref 2. Copyright 2020 Wiley-VCH. (g) Catalytic cycle of multiselective [2 + 2] coupling of acyclic olefins. Adapted with permission from ref 48. Copyright 2020 the authors. Published by Springer Nature under CC BY 4.0 license.

The unconventional selectivities for oxidative 1,4-coupling, Meerwein arylation, and [2 + 2] photocycloaddition reactions at multiple levels have been successfully realized, which are unattainable from molecular catalysts (Figure 15, see more details in supplemental discussion).^{2,43,47,48,53,55}

4.4. Biochemical Applications

The water solubility and hydrophobic cavity suitable for drug loading, as well as the integrated multifunctionalities, make these nanocages promising as biomedical materials in theranostic nanomedicine.^{49–51} For example, the one-/two-photon-excited red-light emissive phosphorescence and pro-

ton-sensitive attributes of multiple $\text{Ru}(\text{ImPhen})_3$ centers in MOC-16- Ru_8Pd_6 make it an ideal pH-dependent imaging probe at both living cell and tissue levels under visible and near-infrared (NIR) excitation.⁵⁰ The electrostatic interactions arising from high-positive charges and the hydrophobicity-lipophilicity balanced via imidazole-deprotonation facilitate its pH-dependent membrane absorption within the biological pH-range, which enables its application in cell membrane tracking of lung tissues under two-photon excitation (Figure 16), showing potential in tumor diagnosis and therapy with low toxicity (Figure S1).

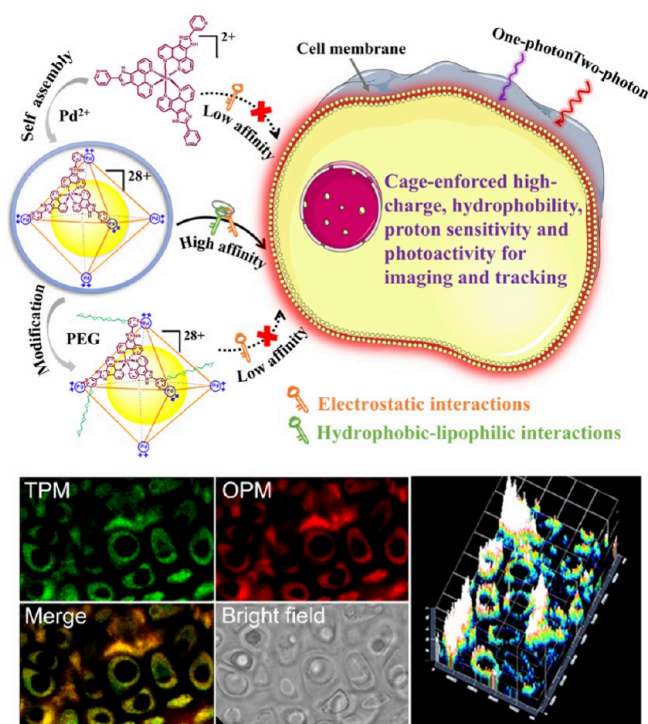


Figure 16. Synergistic effect of MOC-16- Ru_8Pd_6 for membrane absorption and one-/two-photon membrane tracking images of the sectioned lung tissues. Adapted with permission from ref 50. Copyright 2020 ACS.

A more intriguing bioapplication arises from MOC-16- Os_8Pd_6 (MOC-43), which features heavy Os-centers as radiosensitizers and photosensitizers to enable X-ray-induced photodynamic therapy (XPDT) for deep-penetrating cancer treatment.⁴⁹ This Os-cage can directly activate the PDT process at low X-ray dose, efficiently generate singlet-oxygen ($^1\text{O}_2$), and emit NIR-phosphorescence for bioimaging. More-

over, the anticancer drug coumarin can be loaded into the cage, and micellization of the host-guest complexes with DSPE-PEG2000 affords coumarin-MOC-16- Os_8Pd_6 nanoparticles (CMNPs) for the combined treatment of RT-XPDT-Chemotherapies (Figure 17). NIR-imaging reveals that CMNPs localize in lysosomes inside triple-negative breast cancer 4T1 cells, and the enhanced permeability and retention (EPR) effects of CMNPs facilitate efficient tumor targeting, signifying MOC-16- Os_8Pd_6 as an ideal platform for collaborative cancer theranostics.

In vitro studies confirmed that CMNPs exhibit excellent antitumor efficacy with much improved antiproliferative activity against breast cancer 4T1 cells, and the combined therapy of CMNPs is also tested effective in vivo on 4T1 tumor-bearing mice, which results in minimal systemic toxicity (Figure S2) and high anticancer efficacy, underscoring the promising biochemical and biomedical applications of this cage as an integrated solution for cancer diagnosis and treatment (Figure 17).

Very recently, we apply MOC-68 for *in situ* codelivery of H_2S and O_2 gases to inflammatory cell within atherosclerotic plaques, which can be loaded with diallyl trisulfide and doped in PEGylated MnO_2 nanoparticles.⁵¹ In vitro and in vivo studies indicate that this nanodrug holds great promise as a novel theranostic platform for the MRI-visible dual-gas therapy of atherosclerosis.

5. CONCLUSION AND OUTLOOK

By strategically designing and assembling, the intrinsically active $\text{M}(\text{ImPhen})_3$ -centers have been successfully embedded into the MOC structures, which not only bring in various functions including chirality, photoactivity, redox-activity, catalytic-activity, and pH-responsiveness but also showcase biomimetic characteristics like confinement, allosteric, and amphoteric effects. The studies summarized in this Account demonstrate how specific functions of the MOCs could be realized through engineering active cage-nanospaces, even

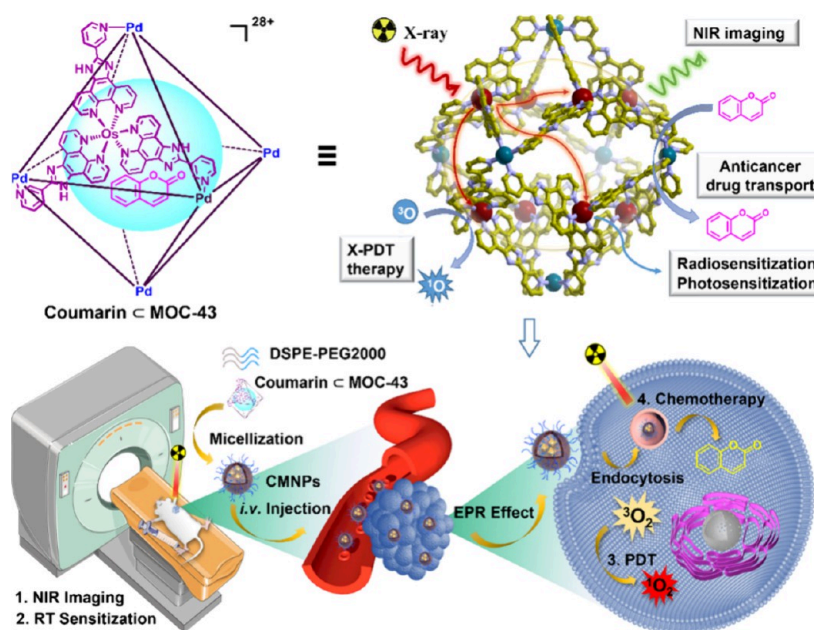


Figure 17. Application of the micellar CMNPs in breast cancer diagnosis and RT-XPDT-Chemotherapy combination. Adapted with permission from ref 49. Copyright 2023 ACS.

more, how to construct an active-MOC for multipurpose applications, and how to achieve enhanced performance through synergizing multifunctionalities in the cage-confined nanospaces.

Looking forward to future perspectives, a promising area of the activated MOCs may be cage-confined catalysis, which imitates but may go beyond enzyme-mimics. The challenging organic transformations with reactivities and selectivities unattainable from conventional molecular catalysts may be expected from cage-catalytic protocol, such as multilevel chemo-, regio-, diastereo-, and enantioselective catalysis. Another but not limited intrigue area may be the cage-based theranostic nanomedicine, which can combine drug delivery, imaging, and multitherapies in a nanocage platform for integrated diagnosis and treatment.

Further efforts on designing active-MOCs for wide and practical applications should be considered to expand the cage structural topology and complexity, e.g., various Platonic and Archimedean polyhedrons with specific cavities, mix-metal/ligand MOCs with more versatile and synergistic functionalities, diversified metal-centers with high/low redox-potentials for ET or excited-states for EnT processes, as well as ultrastable MOCs with organic or kinetically inert metal-vertices against harsh conditions. More cage activation strategies definitely create more possibilities for bottom-up research, laying the groundwork for broad explorations of cage applications.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.accounts.4c00467>.

Supplemental discussion on the anion role in cages, supplemental discussion on electron/energy transfer photocatalysis, and supplemental figures (PDF)

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CRedit: **Yu-Lin Lu** writing - original draft; **Ya-Ping Wang** writing - original draft; **Kai Wu** writing - review & editing; **Mei Pan** writing - review & editing; **Cheng-Yong Su** conceptualization, funding acquisition, project administration, supervision, writing - original draft, writing - review & editing.

Notes

The authors declare no competing financial interest.

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Mei Pan got her Ph.D. degree in Shanghai Institute of Technical Physics, CAS, in 2004. She is now a full professor of chemistry in Sun Yat-sen University, focusing on the study of the design, mechanism, and photoluminescence of metal–organic materials.

Cheng-Yong Su obtained his PhD in 1996 from Lanzhou University. In 2001, he joined Prof. Wolfgang Kaim's group at Stuttgart University as an Alexander von Humboldt Research Fellow, then moved to the University of South Carolina in 2002 to work with Prof. Hans-Conrad zur Loye as a postdoctoral fellow. In 2004, he was promoted to a full professor in Sun Yat-Sen University. His research is interdisciplinary in the field of supramolecular chemistry, materials, and catalysis relevant to clean energy and green transformations.

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